7. Nullspace

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Here is what happens in general for homogeneous linear systems:

Recall that the set of all solutions to a homogeneous linear system $\mathbf{A}\mathbf{x} = \mathbf{0}$ is called the **nullspace of matrix** $\mathbf{A}$, and is denoted $\mathrm{NS}\,(\mathbf{A})$.

Theorem 7.1 If $\mathbf{A}$ is a matrix, then the set of all solutions to the homogeneous linear system $\mathbf{A}\mathbf{x} = \mathbf{0}$ is a vector space. (In other words if $\mathbf{A}$ is an $m \times n$ matrix, then the nullspace $\mathrm{NS}\,(\mathbf{A})$ is a subspace of $\mathbb{R}^n$.)

Analogous theorem : If $P\,(D)$ is a linear differential operator, then the set of solutions to the homogeneous ODE $P\,(D)\,x = 0$ is a vector space of functions. (It is a subspace of the set of all functions.)

Proof of theorem

We already showed that the solutions to $\mathbf{A}\mathbf{x} = \mathbf{0}$ satisfy the definition of a vector space. (In fact we used this to motivate our definition of a vector space.) We replicate the proof here for completeness.

0. $\mathbf{A}\mathbf{0} = \mathbf{0}$ implies that $\mathbf{0}$ is in the nullspace.
1. If $\mathbf{x}$ is in the nullspace, this means that $\mathbf{A}\mathbf{x} = 0$. Then

$$\mathbf{A}\,(c\mathbf{x}) = c\mathbf{A}\mathbf{x} = c\mathbf{0} = \mathbf{0}.$$

Thus $c\mathbf{x}$ is also in the nullspace.
2. If $\mathbf{x}_1$ and $\mathbf{x}_2$ are in the nullspace, this means that $\mathbf{A}\mathbf{x}_1 = 0$ and $\mathbf{A}\mathbf{x}_2 = 0$. Then

$$\mathbf{A}\,(\mathbf{x}_1 + \mathbf{x}_2) = \mathbf{A}\mathbf{x}_1 + \mathbf{A}\mathbf{x}_2 = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Thus $\mathbf{x}_1 + \mathbf{x}_2$ is also in the nullspace.

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Example of nullspace by inspection

5:00 / 5:00 2.0x

The null space is a line.
It's the line through--
it's in R3 and the vector 1, 1, negative 1 maybe
goes down here.
I don't know where it goes, say, down here.
There is the vector 1, 1, negative 1 that you gave me.
And where is the vector C, C negative C?
Its on this line.
Of course, there's 0, 0, 0 that we had.
And what we've got is that whole--
whoops-- the whole line, going both ways, through the origin.
The null space is a line in R3.
For that example, we could find all the combinations

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